

Pyae Phyo Wai

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An Electronic Engineering graduate (First-Class Honours) and Autodesk Certified Professional (ACP) specialising in multi-disciplinary system design across energy, utility, and building services infrastructure. Proven track record executing end-to-end electrical schematics, intelligent 3D modelling, and production-ready manufacturing documentation. Highly proficient across industry-standard engineering software ecosystems, including AutoCAD Electrical, Revit (BIM), SolidWorks, and EPLAN.

Key Technical Skills

- CAD & Electrical Software: AutoCAD Electrical, EPLAN Electric P8 & EPLAN PRO, advanced AutoCAD, external references (Xrefs), scaled layouts.
- 3D Modelling & BIM: Autodesk Revit (BIM coordination & clash detection), SolidWorks (mechanical enclosure prototyping), spatial overlay layouts.
- Core Design & Drafting: Electrical Panel Schematics, Multi-core Wiring Diagrams, Termination Schedules, Network Architecture, Three-Line Diagrams.
- Power Systems & Compliance: Elexon Codes of Practice (CoP1/2/3/5) settlement metering, G99 grid compliance, communication panels, smart-grid monitoring (DSM, OMS, ASM, PQM).
- Engineering Workflows: Drawing quality auditing, Bills of Materials (BOM) compilation, change log management, Red Line markup translation.

Relevant Work Experience

Kenda Electronics Systems LTD., Totton, UK

Dec 2025 to Present

Electrical CAD Designer

- Engineered comprehensive electrical schematics, multi-core wiring diagrams, general arrangement drawings and network architecture diagrams for grid-scale energy projects using AutoCAD.
- Developed production-ready panel layouts for Elexon Code of Practice compliant metering cubicles (CoP1, CoP2, CoP3, CoP5), G99-compliant smart-grid monitoring systems, and dedicated communication panels.
- Utilised SolidWorks to design enclosure mechanical cutout drawings, earthing bar drilling templates, and optimised external cable routing layouts.
- Generated accurate Bills of Materials (BOMs) within database-driven drawing environments to streamline multi-discipline procurement.
- Audited peer engineering drawings for technical accuracy while managing drawing change logs and "As-Built" record tracking using Red Line markups.
- Represented the design engineering team in technical coordination meetings to resolve client queries and incorporate complex site markups.
- Mentored and upskilled trainee technicians in advanced workflows, print reading, and the interpretation of complex client switchgear wiring diagrams.

Aung Tagon Construction Co., LTD., Myanmar

June 2022 to Sept 2024

BIM Modeller/ CAD Designer

- Created and managed highly detailed 2D layouts and coordinated 3D BIM models for complex MEP services (HVAC, plumbing, electrical, and fire protection) using AutoCAD and Revit.
- Translated conceptual designs into precise engineering packages, including General Arrangements (GAs), layouts, and P&ID/SLD schematics, meeting all strict project requirements.
- Integrated and processed LiDAR point cloud scan data to build accurate 3D as-built models for retrofit projects, ensuring precise structural coordination.
- Collaborated independently within a Common Data Environment (CDE) using BIM 360 to manage version control and resolve inter-disciplinary conflicts alongside engineers and architects.
- Supported Project Managers across the project lifecycle by performing clash detection, resolving design challenges, and generating accurate material take-offs and Bills of Materials (BOM).

- Maintained and optimised the company's design libraries by creating new parametric Revit families, developing CAD blocks, and standardising drawing templates to increase team efficiency.

Certifications & Professional Training

Autodesk Certified Professional (ACP) in AutoCAD (2022)

- Produced high-quality 2D and 3D engineering drawings, including General Arrangements (GAs), SLD, and P&ID schematics.
- Mastered advanced drafting techniques to create precise, production-ready manufacturing and installation specifications for the energy and MEP sectors.
- Customised CAD workflows using Sheet Set Management, External References (Xrefs), and dynamic blocks to support automated BOM listings.

Autodesk Certified Professional (ACP) in Revit MEP

- Modelled complex electrical systems, circuits, panel schedules, and intricate piping networks for building energy services.
- Managed advanced BIM coordination processes within a Common Data Environment (CDE), adhering to ISO 19650 standards.
- Executed systematic clash detection to identify and resolve inter-disciplinary design conflicts between MEP, structural, and architectural models.

On-Site MEP Project Engineer Training (Freewill Engineering Training Center)

- Supported the construction phase of complex MEP infrastructure, bridging the gap between 3D BIM models and physical installations.
- Gained practical site experience with electrical, plumbing, sanitary, and firefighting system installations to inform accurate design layouts.
- Reviewed ACMV (HVAC) central plant equipment and distribution networks on-site to ensure design compliance and constructability.

Education

University of Portsmouth, UK

Sept 2024 to June 2025

BEng (Hons) Electronic Engineering (Overall Grade: First-Class)

- Simulated and analysed power conversion and digital control architectures for a solar-powered EV charging station using MATLAB/Simulink and LTSpice to validate performance results.
- Designed analogue and digital electronic circuits, including Chebyshev and Butterworth filters, using schematic capture tools to evaluate transient and frequency responses.
- Developed embedded software in C using FreeRTOS to design and program a smart conveyor system with integrated sensor interfaces.

Chindwin Technological College, Myanmar

June 2021 to June 2023

HND (Level 5) in Electrical & Electronic Engineering (Overall Grade: Merit)

- Designed and built analogue and digital electronic prototypes, utilising hands-on soldering, circuit board assembly, and troubleshooting to evaluate system performance.
- Developed firmware in C for Arduino microcontrollers to manage sensors, automate actuators, and optimise power usage in an Energy Management System.
- Utilised Proteus, KiCAD, and Altium electronic design software to engineer custom PCB layouts featuring both surface-mount (SMD) and through-hole technologies.

Publication and Research

Wai, P.P., Okpako, O., and Anoh, K. (2025). "Design and Simulation of a Solar-Powered EV Charging Station with Battery Backup Using Raspberry Pi Real-Time Monitoring." *IEEE Conference Paper Submission / BEng (Hons) Technical Paper, University of Portsmouth*.

- Authored as the first author to detail the feasibility, design, and modeling of a 4.4kW off-grid solar charging system, PV arrays, battery storage (BESS), and power electronics using MATLAB/Simulink and AutoCAD.

References available on request